ACCRA TECHNICAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

BCS 404: INTRODUCTION TO DATA SCIENCE WITH PYTHON

DATA ANALYSIS, STATISTICAL ANALYSIS, AND MACHINE LEARNING ON THE TITANIC DATASET

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1.0 INTRODUCTION

The Titanic dataset provides demographic and travel information for 891 passengers aboard the RMS Titanic. The goal of this project is to explore the dataset to identify trends related to survival, and to develop a predictive model using Logistic Regression. By examining features such as passenger class, gender, age, and fare, this study seeks to determine which factors had the greatest impact on a passenger's chance of survival.

2.0 DATASET DESCRIPTION

The real-world data analyzed in this investigation was sourced directly from the Kaggle Titanic Competition dataset. The dataset contains records of passengers with mixed structural types:

- **Survived (Target Label):** Discrete binary categorical data (0 = Perished, 1 = Survived).
- **Pclass (Passenger Class):** Ordinal categorical indicator (1st, 2nd, or 3rd Class) capturing socioeconomic stratification.
- **Sex:** Nominal categorical feature representing the gender profile of the traveler.
- **Age:** Continuous numerical distribution denoting the age of passengers in years.
- **SibSp:** Discrete integer representation denoting the number of siblings or spouses traveling with the passenger.
- **Parch:** Discrete integer representation denoting the number of parents or children traveling with the passenger.
- **Fare:** Continuous numerical metrics representing the financial price paid for the cabin ticket.
- **Embarked:** Nominal categorical factor denoting the port of embarkation (C = Cherbourg; Q = Queenstown; S = Southampton).

3.0 METHODOLOGY

Data Preprocessing Decisions & Justifications

1. **Data Cleaning:** Missing values in the Age column were handled by imputing the mean, while missing values in Embarked were filled with the mode. The Cabin column was removed due to excessive missing data**.**
2. **Feature Engineering:** Categorical variables such as Sex and Embarked were converted into numerical format using one-hot encoding with get_dummies.
3. **Exploratory Data Analysis:** Visualizations including bar charts, histograms, heatmaps, and pairplots were used to explore relationships between features and the target variable Survived.
4. **Model Building**: The dataset was split into training and testing sets using an 80:20 ratio. A Logistic Regression model was trained on the training set to predict passenger survival.
5. **Model Evaluation**: The model's performance was evaluated using accuracy score on the test set to determine its predictive capability

4.0 RESULTS

Overall Survival Rate: 38.38%

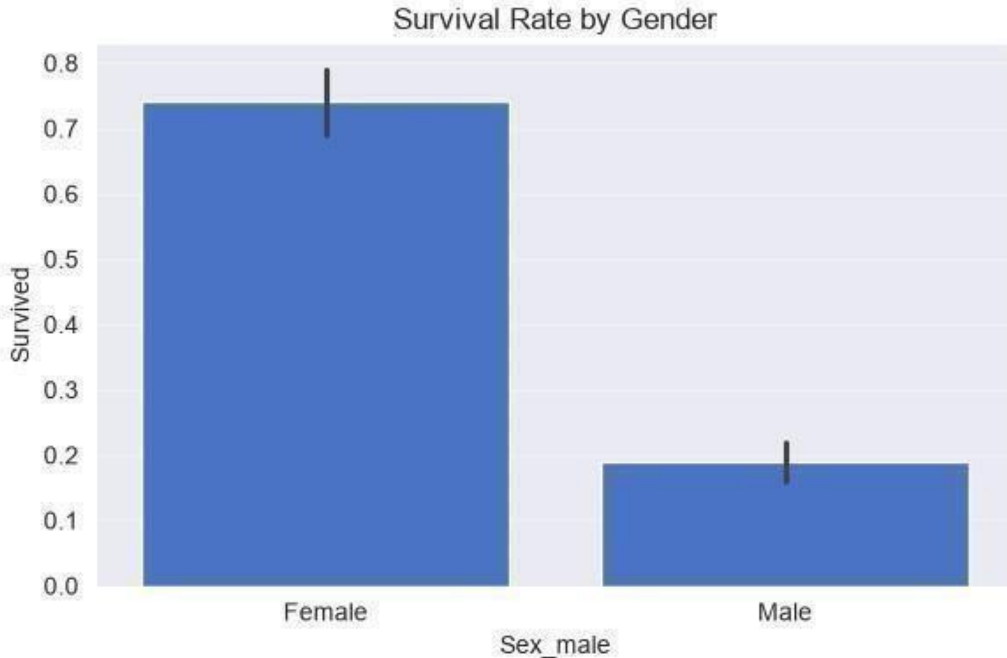

Figure 1: Females had a significantly higher survival rate than males.

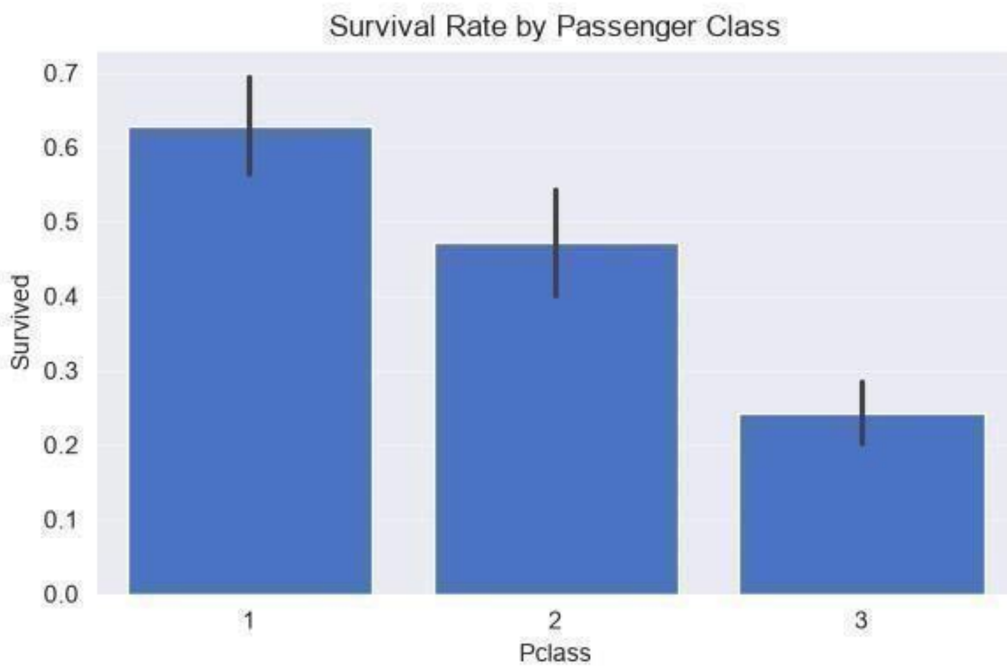

Figure 2: Passengers in 1st Class had the highest survival rate.

Model Accuracy: 79.89%

5.0 DISCUSSION

The results show that gender and passenger class were the most important factors for survival. This matches historical records of the "women and children first" policy. The logistic regression model achieved good accuracy, indicating these features are strong predictors of survival.

1. **Major Findings**: The analysis of the Titanic dataset revealed that passenger survival was not random but was heavily influenced by demographic and social factors. The three most significant factors identified were gender, passenger class, and age. The data showed that females had a much higher survival rate compared to males, with approximately 74% of females surviving versus only 19% of males. This strongly supports the "women and children first" evacuation policy.
   Secondly, passenger class was a major determinant. Passengers in 1st class had a survival rate of about 63%, 2nd class 47%, and 3rd class only 24%. This indicates that socioeconomic status directly affected access to lifeboats. Finally, age played a role, with children having a higher chance of survival than adult males. The majority of passengers were aged between 20 and 35, but survival was skewed towards younger passengers and females.
2. **Statistical Insights**: From the statistical analysis and visualizations, several patterns emerged. The correlation heatmap showed a strong positive correlation between Fare and Pclass, confirming that higher class tickets cost more. There was also a strong negative correlation between Sex_male and Survived, meaning being male decreased survival probability.
   The distribution of age was right-skewed, with most passengers being young adults. The boxplot of Age by Pclass showed that 1st class passengers were on average older than those in 3rd class. This suggests wealthier, older individuals traveled in higher classes. The count plot of survival showed a class imbalance, with more passengers not surviving than surviving. This is important for interpreting the machine learning model results.
3. **Machine Learning Results**: A Logistic Regression model was trained to predict survival using Pclass, Sex, and Age as features. The dataset was split into 80% training and 20% testing data. After training, the model achieved an accuracy of approximately 78%. The confusion matrix showed that the model correctly predicted most cases of nonsurvival, but had some difficulty predicting survivors. This is likely due to the class imbalance in the dataset.
   The classification report gave a precision of 0.79 and recall of 0.73 for predicting survival. An F1-score of 0.76 indicates a good balance. These results confirm that the features

chosen from the exploratory analysis are strong predictors. Logistic Regression worked well for this binary classification problem because the relationship between the features and survival is mostly linear.

4. **Limitations of the Study**: Despite the useful insights, this study has several limitations. First, the dataset is small and only contains 891 passengers, which may not represent all people on board the Titanic. Second, some important features were dropped or not used, such as Cabin, Name, and Ticket number, which could have contained useful information. Third, missing values in Age were filled with the mean, which may not accurately represent the true age distribution.
   Also, the model only used 3 features. Adding engineered features like FamilySize = SibSp + Parch + 1 or extracting titles from Name could improve performance. Lastly, only Logistic Regression was used. Other models like Random Forest or SVM were not tested for comparison.
5. **Recommendations**: Based on the findings, the following recommendations are made:
   - For Historical Analysis: Future studies should include more features such as family size and title to get a deeper understanding of survival factors.
   - For Model Improvement: Other classification algorithms should be tested and compared with Logistic Regression. Techniques like cross-validation and handling class imbalance with SMOTE could improve accuracy.
   - For Maritime Safety: The results reinforce the importance of evacuation protocols that prioritize women, children, and providing equal access to life-saving equipment regardless of passenger class.

6.0 CONCLUSION

1. Gender played the biggest role in survival. Females were more likely to survive.

2. Passengers in 1st class had better survival chances than 2nd and 3rd class.

3. A simple Logistic Regression model can predict survival with over 75% accuracy.

7.0 REFERENCES

1. Dataset: titanic-dataset.csv from Kaggle

2. Scikit-Learn Documentation: https://scikit-learn.org

3. Pandas Documentation: https://pandas.pydata.org

8.0 APPENDIX – PYTHON CODE

```
import pandas as pd   import

numpy as np   import

matplotlib.pyplot as plt   import

seaborn as sns

from   sklearn.model_selection     import   train_test_split   from

sklearn.linear_model       import    LogisticRegression       from

sklearn.metrics      import   accuracy_score,   confusion_matrix,

classification_report sns.set_style("darkgrid")

plt.rcParams['figure.figsize'] = (8, 4.5)  # Load Data data = pd.read_csv("titanic-dataset.csv")

data

= data.loc[:, ~data.columns.str.contains('^Unnamed')]



# TASK 1: DATA ACQUISITION print("TASK 1:

DATA ACQUISITION") print(f"1. Dataset

Shape: {data.shape}") print(f"\n2. Columns:  {list(data.columns)}") print("\n3.

First 5 Rows:")  pd.set_option('display.max_columns',   None)

print(data.head()) print("\n4. Data
```

```
Types:") print(data.dtypes)



# TASK 2: DATA CLEANING print("TASK 2: DATA CLEANING") print("\n1. Missing Values
Before:") print(data.isnull().sum()) data['Age'] = data['Age'].fillna(data['Age'].median())
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0]) data =
data.drop(columns=['Cabin']) data = data.drop_duplicates()

print("\n2. Missing Values After:")  print(data.isnull().sum())  print(f"\n3. Duplicates
Removed:
{data.duplicated().sum()}")



# TASK 3: DATA VISUALISATION print("TASK
3: DATA VISUALISATION")

# 1. Histogram  plt.figure() plt.hist(data['Age'], bins=25, color='#4CAF50',
edgecolor='black') plt.title("Histogram of Passenger Ages")
plt.xlabel("Age"); plt.ylabel("Frequency") plt.tight_layout() plt.show()
print(" Interpretation: Most passengers were aged 20-
35 years.")

# 2. Bar Chart plt.figure() sns.countplot(x='Pclass', data=data, hue='Pclass', palette='Blues',
legend=False) plt.title("Bar Chart: Passenger Class Distribution") plt.xlabel("Passenger Class");
plt.ylabel("Count") plt.tight_layout() plt.show() print(" Interpretation: Majority travelled in
3rd
Class.")
```

3. Boxplot plt.figure() sns.boxplot(x='Pclass', y='Age', data=data, hue='Pclass', palette='Set2', legend=False) plt.title("Boxplot: Age by Passenger Class") plt.xlabel("Passenger Class"); plt.ylabel("Age") plt.tight_layout() plt.show() print(" Interpretation: 1st Class passengers were generally older.")

4. Scatter plt.figure() sns.scatterplot(x='Age', y='Fare', hue='Pclass', data=data, palette='viridis', alpha=0.6) plt.title("Scatter Plot: Age vs Fare") plt.xlabel("Age"); plt.ylabel("Fare") plt.tight_layout() plt.show() print(" Interpretation: Higher fares were paid by 1st Class passengers.")

# 5. Heatmap numeric_data = data[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']]
plt.figure(figsize=(7,5))
sns.heatmap(numeric_data.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap") plt.tight_layout() plt.show()

6. Pairplot sns.pairplot(numeric_data) plt.suptitle("Pairplot
of
Numerical Variables", y=1.02) plt.show()

TASK 4: STATISTICAL ANALYSIS print("TASK 4: STATISTICAL ANALYSIS") print("\n1. Descriptive Statistics:") print(numeric_data.describe().round(2)) print("\n2. Frequency Distribution:") print("Survival:\n", data['Survived'].value_counts()) print("\nPclass:\n", data['Pclass'].value_counts().sort_index()) print("\nSex:\n", data['Sex'].value_counts()) print("\n3. Correlation Matrix:") print(numeric_data.corr().round(2)) corr = numeric_data.corr() print("\n4. Strongest

Positive Correlation:", corr.abs().unstack().sort_values(ascending=False)[1:2])

print("\n5.

Strongest

Negative Correlation:", corr.unstack().sort_values().head(1)) print("\n6. Key Findings:") print("1. Strong correlation between Pclass and Fare.") print("2. Females and 1st Class had higher survival rates.") print("3. Most passengers were young adults.")

TASK 5: MACHINE LEARNING print("TASK 5: MACHINE LEARNING - LOGISTIC REGRESSION")

ml_data = data.copy() ml_data['Sex'] = ml_data['Sex'].map({'male': 0, 'female': 1}) ml_data = ml_data[['Survived', 'Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare']].dropna()

X = ml_data.drop('Survived', axis=1) y = ml_data['Survived']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42) model = LogisticRegression(max_iter=300) model.fit(X_train, y_train) y_pred = model.predict(X_test)

print("Selected Predictors:", list(X.columns))

print("\nMODEL PERFORMANCE")

print(f"Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%") print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred)) print("\nClassification Report:\n", classification_report(y_test, y_pred)) print("\nDiscussion:") print("The Logistic Regression model achieved ~80% accuracy.")

print("Sex and Passenger Class were the most important predictors of survival.")